

Multiplicative structure of $\mathbb{Q}_p^*$

We denote by $\mathbb{Q}_p^*$ the group of non-zero elements of the field $\mathbb{Q}_p$.

Given a non-zero p -adic number x , we have $v_p(x) = k$ for some integer k . It follows that $x = p^k u$ where u is a p -adic number with $v_p(u) = 0$. So u is in $\mathbb{Z}_p$. Now, $v = 1/u$ also satisfies $v_p(v) = 0$, so it is also an element of $\mathbb{Z}_p$.

Every non-zero p -adic number is of the form $p^k u$ where u is an invertible element of $\mathbb{Z}_p$.

The group $U(\mathbb{Z}_p)$ of invertible elements of $\mathbb{Z}_p$ is called the group of p -adic units. As seen above, it is precisely the subset of $\mathbb{Q}_p$ consisting of elements u such that $v_p(u) = 0$.

We have an isomorphism $\mathbb{Z} \times U(\mathbb{Z}_p) \rightarrow \mathbb{Q}_p^*$ given by $(k, u) \mapsto p^k u$.

Thus, in order to understand the structure of $\mathbb{Q}_p^*$ it is enough to understand the structure of $U(\mathbb{Z}_p)$. To do this we need some facts about power series in $\mathbb{Q}_p$.

Power Series in $\mathbb{Q}_p$.

A fundamental fact about p -adic series $\sum_{n=0}^{\infty} x_n$ is as follows:

A series $\sum_{n=0}^{\infty} x_n$ of p -adic numbers converges if and only if (x_n) converges to 0 in the p -adic norm.

Recall that convergence of the series is the same as the convergence of the sequence (s_m) of partial sums $s_m = \sum_{n=0}^m x_n$.

Thus, if the series converges, given any integer k , there is an N such that, for all $m, n \geq N$ we have $|s_m - s_n|_p < 1/k$. Since $x_n = s_{n+1} - s_n$, we see that $|x_n|_p < 1/k$ for $n \geq N$. This shows that (x_n) converges to 0.

Conversely, if (x_n) converges to 0, then for any integer k , there is an N such that, for all $n \geq N$, we have $v_p(x_n) > k$. For $m \geq n \geq N$, we have $s_m - s_n = \sum_{j=n+1}^m x_j$, so we have

$$v_p(s_m - s_n) \geq \min\{v_p(x_j) : n < j \leq m\} > k$$

This shows that (s_n) is a Cauchy sequence.

We now apply this to study units in $\mathbb{Z}_p$.

Geometric Series

For x in $p\mathbb{Z}_p$, we have $v_p(x^k) \geq k$. Thus, (x^n) converges to 0. It follows that $\sum_{k=0}^{\infty} x^k$ converges. Moreover, since the partial sums give a sequence in $\mathbb{Z}_p$, the limit is also in $\mathbb{Z}_p$.

In fact, note that $\sum_{k=0}^{\infty} x^k = 1 + x \sum_{k=0}^{\infty} x^k$ is in $1 + p\mathbb{Z}_p$. Re-writing this, we see that

$$(1 - x) \sum_{k=0}^{\infty} x^k = 1$$

Thus we see that $\sum_{k=0}^{\infty} x^k$ is the inverse of $1 - x$.

If $y = 1 - x$ lies in $1 + p\mathbb{Z}_p$, then its inverse is given by $\sum_{k=0}^{\infty} x^k$ and it also lies in $1 + p\mathbb{Z}_p$.

This proves that $1 + p\mathbb{Z}_p$ is contained in the unit group $U(\mathbb{Z}_p)$.

If $v_p(x) \geq 1$ and $v_p(y) \geq 1$, then $v_p(x + y + x \cdot y) \geq 1$. So, $1 + p\mathbb{Z}_p$ is closed under multiplication. As seen above, it is also closed under taking inverses. Thus, it is a subgroup of $U(\mathbb{Z}_p)$.

The same argument can be applied to $U_r(\mathbb{Z}_p) = 1 + p^r\mathbb{Z}_p$ for any $r \geq 1$ to show that it is a subgroup of $U(\mathbb{Z}_p)$. Thus, we have a descending chain of subgroups

$$U(\mathbb{Z}_p) \supset U_1(\mathbb{Z}_p) \supset U_2(\mathbb{Z}_p) \subset \dots$$

Using the p -adic expansion of elements of $\mathbb{Z}_p$, we see that:

- U/U_1 can be identified with the set $\{a : 0 < a < p\}$ which is the set of invertible elements in $\mathbb{Z}/\langle p \rangle$.
- U_r/U_{r+1} can be identified with the set of elements $\{(1 + p^r a) : 0 \leq a < p\}$ which is a group of invertible elements in $\mathbb{Z}/\langle p^{r+1} \rangle$.

For $r \geq 1$, we see that $(1 + p^r a)(1 + p^r b) = 1 + p^r(a + b)$ in $\mathbb{Z}/\langle p^{r+1} \rangle$, since $2r \geq r + 1$. Thus, U_r/U_{r+1} can be identified with the *additive* group $\mathbb{Z}/\langle p \rangle$.

On the other hand U/U_1 is clearly identified with the multiplicative group of non-zero elements of the field $\mathbb{Z}/\langle p \rangle$.

Roots of unity

A non-zero element a in $\mathbb{Z}/\langle p \rangle$ satisfies $a^{p-1} = 1$. Thus, it is a $(p - 1)$ -th root of unity in this field.

It follows that $a^p = a$ in $\mathbb{Z}/\langle p \rangle$. Thus, for a in $\{0, 1, \dots, p - 1\}$, we see that $a^p - a$ is divisible by p .

Given *any* $u \in U(\mathbb{Z})$, we can write $u = a + px$ for some $x \in \mathbb{Z}_p$ and some a in $\{0, 1, \dots, p - 1\}$. Hence, $u^p = a^p + py$ by an application of the Binomial theorem. Since $a^p - a$ is divisible by p , we see that $u^p = u + pz$ for some z in $\mathbb{Z}_p$.

Suppose $v^p = v + p^k z$ for v in $U(\mathbb{Z}_p)$ and z in $\mathbb{Z}_p$ and $k \geq 1$. We calculate

$$(v + p^k z)^p = v^p + \sum_{r=1}^{p-1} \binom{p}{r} p^{kr} z^r + p^{kp} z^p$$

For $1 \leq r \leq p-1$, we see that p does not divide $r!$ or $(p-r)!$. It follows that p divides $\binom{p}{r}$ for $1 \leq r \leq p-1$. Thus, the terms in the middle are all divisible by at least p^{k+1} . Since $p \geq 2$ we see that $kp \geq 2k \geq k+1$. Hence, $(v + p^k z)^p = v^p + p^{k+1} z'$ for some z' in $\mathbb{Z}_p$. This shows that $v_p(v^{p^2} - v^p) > v_p(v^p - v)$. Hence,

For $u \in U(\mathbb{Z}_p)$, the sequence (u^{p^n}) converges in $\mathbb{Z}_p$.

Note that if $\omega(u)$ is the limit of this sequence, then $\omega(u) - u$ lies in $p\mathbb{Z}_p$. Thus, $\omega(u)$ is in $u + p\mathbb{Z}_p$ as is a unit in $\mathbb{Z}_p$.

Moreover, taking p -th powers, we see that $(u^{p^n})^p = u^{p^{n+1}}$. Thus, taking limits of both sides as n goes to infinity, we see $\omega(u)^p = \omega(u)$. It follows that $\omega(u)^{p-1} = 1$.

For every $u \in U(\mathbb{Z}_p)$, the element $\omega(u)$ is a $(p-1)$ -th root of unity such that $v_p(u - \omega(u)) > 1$.

In other words, we have found a subgroup of $U(\mathbb{Z}_p)$ which maps *isomorphically* onto the group of units in $\mathbb{Z}/\langle p \rangle$. (Since the group of q -th roots of unity in a field of characteristic 0 is a cyclic group of order q , we see that the units in $\mathbb{Z}/\langle p \rangle$ forms a cyclic group of order $(p-1)$.)

We note that if $v_p(u - v) > 1$, then $\omega(u) = \omega(v)$. Thus, by abuse of notation, we can also write $\omega(a)$ for a in $(\mathbb{Z}/\langle p \rangle)^*$.

We have an isomorphism $(\mathbb{Z}/\langle p \rangle)^* \times U_1(\mathbb{Z}_p) \rightarrow U(\mathbb{Z}_p)$ given by $(a, u) \mapsto \omega(a) \cdot u$.

One point worth noting is that this statement has no content when $p = 2$! In that case, we have $U(\mathbb{Z}_2) = U_1(\mathbb{Z}_2)$ and we note:

We have an isomorphism $\{\pm 1\} \times U_2(\mathbb{Z}_2) \rightarrow U(\mathbb{Z}_2)$ given by $(s, u) \mapsto s \cdot u$.

Exponential

Consider the *formal* series $\exp(x) = \sum_{k=0}^{\infty} x^k/(k!)$. We note that

$$v_p(k!) \leq \frac{k}{p} + \frac{k}{p^2} + \cdots = k \frac{\left(\frac{1}{p}\right)}{\left(1 - \frac{1}{p}\right)} = \frac{k}{p-1}$$

In particular, we see that:

- if $p > 2$ and $v_p(x) \geq 1$, so that $v_p(x^k/(k!)) \geq k - k/(p-1)$, or
- if $p = 2$ and $v_2(x) \geq 2$, so that $v_2(x^k/(k!)) \geq 2k - k$

then $v_p(x^k/(k!))$ goes to infinity as k goes to infinity.

Thus, the series $\sum_{k=0}^{\infty} x^k/(k!)$ converges for x in $p\mathbb{Z}_p$ for $p > 2$ and in $4\mathbb{Z}_2$ for $p = 2$, and defines a function pexp from $p\mathbb{Z}_p$ to $U_1(\mathbb{Z}_p)$ for $p > 2$ and from $4\mathbb{Z}_2$ to $U_2(\mathbb{Z}_2)$ when $p = 2$.

We further check that:

- if $p > 2$ and $v_p(x) = k \geq 1$, then $\text{pexp}(x)$ is in $1 + x + p^{k+1}\mathbb{Z}_p$, and
- if $p = 2$ and $v_2(x) = k \geq 2$, then $\text{pexp}(x)$ is in $1 + x + 2^{k+1}\mathbb{Z}_p$.

This shows that $\text{pexp}(x) \neq 1$ for all $x \neq 0$.

One can check the following *formal* identity (using the Binomial Theorem!):

$$\sum_{k=0}^{\infty} \frac{(x+y)^k}{k!} = \left(\sum_{k=0}^{\infty} \frac{x^k}{k!} \right) \cdot \left(\sum_{k=0}^{\infty} \frac{y^k}{k!} \right)$$

It follows that pexp is an injective group homomorphism.

Now, consider the *formal* series $\log(1+x) = -\sum_{k=1}^{\infty} (-x)^k/k$. When $v_p(x) \geq 1$, we see that $v_p((-x)^k/k) \geq k - v_p(k)$. Since the largest power of p that can divide k is at most $\log_p(k)$ (usual logarithm!), we see that $v_p((-x)^k/k) \geq k - \log_p(k)$. One easily shows that this goes to infinity as k goes to infinity.

Thus, the series $\log(x) = \log(1 - (1-x)) = \sum_{k=1}^{\infty} (1-x)^k/k$ converges for x in $U_1(\mathbb{Z}_p)$ and defines a function $\text{plog} : 1 + p\mathbb{Z}_p \rightarrow p\mathbb{Z}_p$.

Now, $\exp(\log(x)) = x$ can be checked since both series converge for sufficiently small *real* (or *complex*!) values of x . This shows that the *same* identity holds for the *formal* series since this identity only involves *finitely many* terms on both sides for each power of x .

It follows that the map $\text{pexp} \circ \text{plog}$ is the *identity* map wherever both maps are defined. This proves that pexp is *surjective* on $U_1(\mathbb{Z}_p)$ for $p > 2$ and $U_2(\mathbb{Z}_2)$ for $p = 2$.

In other words, we see that pexp defines a *group isomorphism* from $p\mathbb{Z}_p$ to $U_1(\mathbb{Z}_p)$ for $p > 2$ and from $4\mathbb{Z}_2$ to $U_2(\mathbb{Z}_2)$ for $p = 2$.

Summary

Putting all of the above together, we get group isomorphisms

$$\begin{aligned} \mathbb{Z} \times (\mathbb{Z}/\langle p \rangle)^* \times p\mathbb{Z}_p &\rightarrow \mathbb{Q}_p^* \text{ for } p > 2 \\ \mathbb{Z} \times \{\pm 1\} \times 4\mathbb{Z}_2 &\rightarrow \mathbb{Q}_2^* \text{ for } p = 2 \end{aligned}$$

given by $(k, a, x) \mapsto p^k \omega(a) \text{pexp}(x)$ in the first case and by $(k, s, x) \mapsto 2^k s \text{pexp}(x)$ in the second case.

These isomorphisms are analogous to the isomorphism $\{\pm 1\} \times \mathbb{R} \rightarrow \mathbb{R}^*$ given by $(s, x) \mapsto s \exp(x)$ for real numbers.